

A Coulomb Route to Solar Fusion:

$$p + e + p \rightarrow d \text{ and } d + d \rightarrow {}^4\text{He}$$

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Physical argument by the author; numerical implementation with AI assistance (Claude, Anthropic).

Abstract

In the standard proton–proton chain, solar hydrogen burning begins with $p + p \rightarrow d + e^+ + \nu_e$, a *weak*-force reaction that converts a proton into a neutron with emission of a positron and a neutrino, and is the slow, rate-limiting step of the whole chain. In the RealNucleus picture — in which the neutron is a bound proton–electron pair and the deuteron is $2p + 1e$ — the same first step is purely *electromagnetic*: $p + e + p \rightarrow d$, an electron from the plasma captured as the deuteron’s glue, with no flavour change, no positron, and no neutrino. Two deuterons then fuse to the alpha, $d + d \rightarrow {}^4\text{He}$, a reaction computed directly in RealQM. The net process $4p + 2e \rightarrow {}^4\text{He}$ releases the same ~ 26.7 MeV as the standard chain but retains the ~ 0.5 MeV per alpha ($\sim 2\%$) that the standard chain loses to neutrinos. The decisive difference is a lepton one: in RealNucleus no lepton is created or destroyed — the electron persists as the deuteron’s constituent — so lepton-number conservation requires *no* neutrino, whereas the standard neutrino is the signature of the flavour-changing weak vertex $p \rightarrow n$. We set the two schemes side by side and identify the experimentum crucis: the solar neutrino flux, directly measured, which the standard weak picture predicts and which the Coulomb picture must account for. The comparison is thus sharp and falsifiable.

Keywords: solar fusion; proton–proton chain; deuteron formation; electron capture; RealNucleus; weak force; solar neutrinos; hydrogen burning.

1 Introduction

The Sun shines by fusing hydrogen to helium, $4p \rightarrow {}^4\text{He}$, releasing 26.73 MeV per helium nucleus. In the standard account [1, 2] this proceeds through the proton–proton (pp) chain, whose first and rate-limiting step,

$$p + p \longrightarrow d + e^+ + \nu_e, \tag{1}$$

is a *weak*-interaction process: two protons collide, one is converted to a neutron ($p \rightarrow n + e^+ + \nu_e$), and the resulting proton and neutron bind as a deuteron. Because it requires the weak force acting during the brief instant two protons are within nuclear range, this step is extraordinarily slow — it is what makes the Sun’s hydrogen last billions of years — and it emits a neutrino that escapes the Sun directly.

RealNucleus [5, 6] offers a different account of the same burning. There the neutron is not elementary but a bound proton–electron pair, $n = (p e^-)$, and the deuteron is accordingly $2p + 1e$: two protons glued by a single shared electron, the charge-conjugate of the one-electron molecular ion H_2^+ . In this picture the first step of hydrogen burning is not weak but electromagnetic,



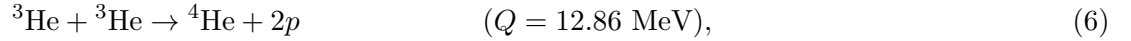
an electron drawn from the surrounding plasma and captured as the deuteron’s glue — no proton is converted, no positron is created, and no neutrino is required. The chain then closes with the direct fusion of two deuterons,



a reaction we have computed directly in RealQM [5]. This paper places the two accounts side by side, compares their energetics and their lepton content, and identifies the single measurement that decides between them.

2 The standard proton–proton chain

The dominant (ppI) branch of the standard chain is



with the two positrons annihilating on plasma electrons. Summing the branch and the annihilations, the net is



of which the two neutrinos carry away, on average, ~ 0.5 MeV, so that ~ 26.2 MeV is deposited as heat. The rate is set by two factors in series: the Coulomb barrier the two protons must tunnel (the Gamow factor, steeply temperature-dependent), *and* the weak matrix element for $p \rightarrow n$ conversion during the collision. The second factor is tiny, and it is the weak bottleneck — not the Coulomb barrier alone — that fixes the Sun’s luminosity and lifetime. A minor variant, the *pep* reaction $p + e^- + p \rightarrow d + \nu_e$ (about 0.4% of pp), captures a plasma electron instead of emitting a positron, but is still weak and still emits a neutrino.

3 The Coulomb route

In RealNucleus the ground state of a nucleus is a set of proton and electron charge clouds on non-overlapping domains with free boundaries, minimising the Coulomb (electrostatic) energy alone,

$$E = \sum_i \frac{1}{2} \int |\nabla \psi_i|^2 + \frac{1}{2} \iint \frac{\rho(x)\rho(y)}{|x-y|} dx dy, \quad \rho = \sum_i q_i \psi_i^2, \quad (8)$$

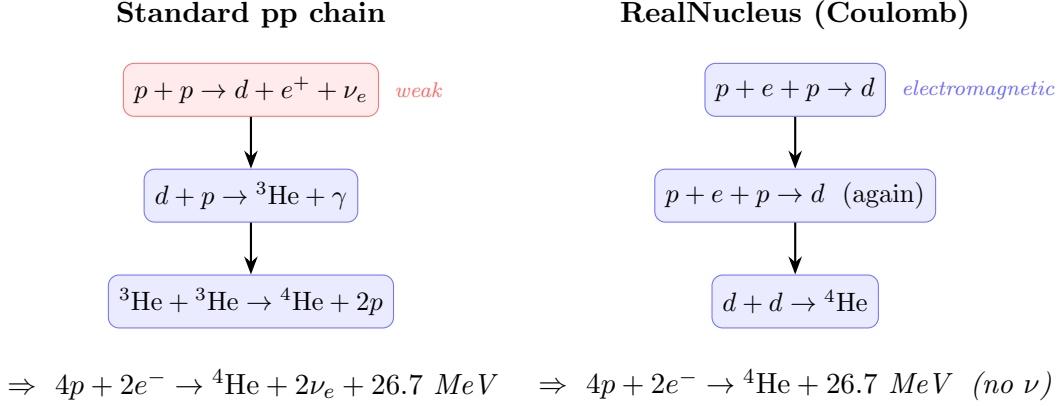


Figure 1: The two accounts of solar hydrogen burning. *Left:* the standard pp chain opens with the weak reaction $p + p \rightarrow d + e^+ + \nu_e$ (red), which converts a proton to a neutron and emits a neutrino. *Right:* the RealNucleus chain opens with the electromagnetic capture $p + e + p \rightarrow d$ (blue), the electron becoming the deuteron’s glue, and closes with the directly-computed $d + d \rightarrow {}^4\text{He}$. Both net to $4p + 2e^- \rightarrow {}^4\text{He}$ at 26.7 MeV; only the standard chain emits neutrinos.

with $q_i = \pm 1$ [6, 5]. A free proton and a free electron form the model neutron $1p + 1e$, which does *not* bind on its own (matching the instability of the free neutron); but a *second* proton produces the deuteron $2p + 1e$, which does. The single electron cannot glue one proton into a bound unit yet glues two — the charge-conjugate $\text{H}^- / \text{H}_2^+$ configuration, the two protons splitting into halfspaces around the central electron core. This is the reaction (2): it needs only an electron, always abundant in the solar plasma, and the Coulomb attraction that binds it; no proton is transmuted.

The second step (3), $d + d \rightarrow {}^4\text{He}$, is the fusion of two such units into the alpha $4p + 2e$ — the closed, dual-confined structure in which a -2 electron core is caged by four protons. In RealQM this reaction is computed directly: two deuterons, each a central electron core with its two protons split into halfspaces, are brought together and relaxed, and the energy falls monotonically onto the deeply bound alpha as the separation decreases [5]. The overall balance is

$$4p + 2e^- \longrightarrow {}^4\text{He} + 26.73 \text{ MeV}, \quad (9)$$

identical in energy to the standard net (7) except that no energy is carried off by neutrinos.

Which second step? A caveat on abundances is in order. In the solar core protons outnumber deuterons enormously: a deuteron, once formed, is consumed almost at once, so its equilibrium abundance is only $[d]/[p] \sim 10^{-18}$. In the standard chain this is precisely why deuterium burns by $d + p \rightarrow {}^3\text{He}$ and not by $d + d$, which is utterly negligible. The same abundance argument applies to the Coulomb route: the dominant second step in a proton-rich plasma is likewise $d + p \rightarrow {}^3\text{He}$ — in RealNucleus simply $3p + 1e$, again electromagnetic — followed by ${}^3\text{He} + {}^3\text{He} \rightarrow {}^4\text{He} + 2p$, exactly the ppI topology but with every step Coulombic and neutrino-free. We write the chain as $d + d \rightarrow {}^4\text{He}$ in (3) because that reaction is the one computed directly in RealQM [5] and the cleanest picture of two dual-confined units merging into the closed alpha; the abundance-weighted path through ${}^3\text{He}$ tells the same electromagnetic story, $4p + 2e^- \rightarrow {}^4\text{He}$, and is equally free of the

weak vertex. The essential contrast with the standard chain therefore lies entirely in the *first* step, the formation of the deuteron: whether it is the weak $p + p \rightarrow d + e^+ + \nu_e$ or the electromagnetic $p + e + p \rightarrow d$. Every step thereafter is Coulombic in both accounts; only the entry point carries — or does not carry — the neutrino.

4 Comparison

Table 1 sets the two accounts against each other. They agree on the overall transmutation and on its energy release; they differ on the *mechanism* of the first step, and hence on the leptons and the neutrinos.

	Standard pp chain	RealNucleus (Coulomb)
first step	$p + p \rightarrow d + e^+ + \nu_e$	$p + e + p \rightarrow d$
force	weak (flavour-changing $p \rightarrow n$)	electromagnetic (Coulomb)
what the deuteron is	$p + n$, n elementary	$2p + 1e$, $n = (pe^-)$ bound
second step	$d + p \rightarrow {}^3\text{He}$, then ${}^3\text{He} + {}^3\text{He}$	$d + d \rightarrow {}^4\text{He}$ (computed)
leptons	e^+ and ν_e created	electron retained; none created
neutrinos	yes ($2\nu_e$ per ${}^4\text{He}$)	none
net	$4p + 2e^- \rightarrow {}^4\text{He} + 2\nu_e$	$4p + 2e^- \rightarrow {}^4\text{He}$
energy	26.73 MeV (~ 0.5 to ν)	26.73 MeV (all retained)
rate limiter	weak vertex $\times$ Coulomb barrier	Coulomb barrier (three-body pep)

Table 1: The standard proton–proton chain and the RealNucleus Coulomb route to solar hydrogen burning, step by step.

The lepton argument. The cleanest distinction is one of lepton bookkeeping. In the standard first step a proton is turned into a neutron, a flavour-changing weak transition that must create a positron and a neutrino to conserve charge and lepton number: the neutrino is the *signature of the weak vertex*. In the Coulomb route no proton is transmuted and no lepton is created or destroyed — the plasma electron simply becomes the deuteron’s glue and remains as the constituent electron of the $n = (pe^-)$ pair. Lepton number is then conserved trivially, with no neutrino. The presence or absence of the solar neutrino is thus a direct readout of which mechanism operates.

5 Energetics and the neutrino budget

Both routes conserve energy at the same value because both effect the same net transmutation $4p + 2e^- \rightarrow {}^4\text{He}$, whose Q -value is fixed by the masses,

$$Q = (4m_p + 2m_e - m_{{}^4\text{He}})c^2 = (3753.09 + 1.02 - 3727.38) \text{ MeV} = 26.73 \text{ MeV}. \quad (10)$$

The standard chain gives up about 0.5 MeV of this to its two neutrinos (mean pp-neutrino energy ≈ 0.27 MeV each), so ≈ 26.2 MeV becomes solar heat and light; the Coulomb route retains the full 26.73 MeV. Equivalently, the standard model predicts a solar *neutrino luminosity* of about 2% of the total, while the Coulomb route predicts essentially none. This few-percent split is small in the energy budget but it is the whole of the observable neutrino signal.

6 The decisive test: solar neutrinos

Here the comparison becomes an experiment, and honesty requires stating the current evidence plainly. The standard chain predicts a pp-neutrino flux at Earth of about $6 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$, tied directly to the solar luminosity, and solar neutrinos are *observed*: the chlorine and gallium radiochemical experiments, Super-Kamiokande and SNO (which established flavour oscillation and measured the total ^8B flux), and above all Borexino, which has directly measured the low-energy pp, ^7Be and pep neutrinos in real time at fluxes consistent with the standard solar model [4, 3, 2]. Detection is by neutrino–electron scattering and inverse beta decay, which count arriving neutrinos in a largely production-model-independent way. Taken at face value, this flux is strong evidence for a neutrino-emitting — hence weak — first step, and it is the principal challenge the Coulomb route must meet: a strictly neutrino-free burning is disfavoured by the measured flux unless that flux can be attributed to another source.

We therefore state the position without overreach. The Coulomb route is a well-defined, computable, and energetically exact account of $4p + 2e^- \rightarrow ^4\text{He}$ that removes the awkward features of the weak first step — the unbound diproton, the improbable in-flight $p \rightarrow n$ conversion — and predicts a specific, falsifiable consequence: little or no solar neutrino emission from the initial deuteron formation. The measured pp-neutrino flux is the experimentum crucis. If that flux is genuinely of primary solar origin at the standard-model value, the strictly neutrino-free version is excluded and any Coulomb account must incorporate a neutrino-producing channel; if some part of the inferred flux rests on assumptions about the production mechanism, that part is open to reexamination. Either way the question is settled by measurement, not by preference — which is the proper standing of a physical proposal.

7 Rates and the Coulomb barrier

Both routes share the Coulomb barrier: two protons must tunnel through their mutual repulsion to reach the range at which an electron can glue them, so both inherit the steep Gamow temperature dependence that localises hydrogen burning to the solar core. The routes differ in what multiplies that barrier. The standard rate carries the additional tiny weak matrix element, which is the dominant suppression. The Coulomb route replaces it with the phase-space factor for a *three-body* encounter $p + e + p$ — the requirement that an electron be present in the small volume where two protons briefly meet — which in a dense plasma is far less suppressed than the weak vertex. The two pictures thus make *different rate predictions* for a given core temperature and density: whether the observed solar burning rate is better matched by the weak bottleneck or by the three-body Coulomb capture is a second, quantitative discriminator, to be developed with a proper plasma

phase-space calculation alongside the neutrino test.

8 Conclusion

Solar hydrogen burning can be told two ways. In the standard proton–proton chain the first step $p + p \rightarrow d + e^+ + \nu_e$ is a weak, neutrino-emitting conversion of a proton into a neutron. In the RealNucleus picture the neutron is a bound proton–electron pair, so the same step is the electromagnetic capture $p + e + p \rightarrow d$, followed by the directly computed fusion $d + d \rightarrow {}^4\text{He}$; no proton is transmuted, no lepton is created, and no neutrino is required. Both net to $4p + 2e^- \rightarrow {}^4\text{He}$ at 26.73 MeV, differing only in the $\sim 2\%$ the standard chain gives to neutrinos. The accounts are therefore not vague alternatives but sharply opposed at one observable: the solar neutrino flux. The standard chain predicts it and it is measured; the Coulomb route predicts its absence from the primary step. We put the two side by side precisely so that the measurement can decide.

Code and simulations

The $d + d \rightarrow {}^4\text{He}$ fusion and the deuteron and alpha structures are computed with an open-source WebGPU RealQM solver, runnable from the interactive gallery [7] at <https://claes542.github.io/RealMolecule/gallery.html>; see in particular the deuteron ([nucleus_2h.html](#)), the alpha ([nucleus_he4_2plus2.html](#)), and the D + D fusion sweep ([nucleus_DD_fusion_Rsweep.html](#)).

Acknowledgements

The physical argument and the model are the author’s; the numerical implementation was developed and run with AI assistance (Claude, Anthropic).

References

- [1] H. A. Bethe, *Energy Production in Stars*, Phys. Rev. **55**, 434 (1939).
- [2] E. G. Adelberger *et al.*, *Solar fusion cross sections. II. The pp chain and CNO cycles*, Rev. Mod. Phys. **83**, 195 (2011).
- [3] Q. R. Ahmad *et al.* (SNO Collaboration), *Direct Evidence for Neutrino Flavor Transformation from Neutral-Current Interactions in the Sudbury Neutrino Observatory*, Phys. Rev. Lett. **89**, 011301 (2002).
- [4] M. Agostini *et al.* (Borexino Collaboration), *Comprehensive measurement of pp-chain solar neutrinos*, Nature **562**, 505 (2018).
- [5] C. Johnson, *RealNucleus: Nuclear Binding as Dual Coulomb Confinement, without a Strong or Weak Force* (companion paper).
- [6] C. Johnson, *RealQM: A Real Quantum Mechanics*, <https://realqm.blogspot.com>.

- [7] C. Johnson, *RealMolecule / RealQM interactive simulation gallery*, <https://claes542.github.io/RealMolecule/gallery.html>; source <https://github.com/Claes542/RealMolecule>.